Jianan Chen

Ph.D. in Computer Science | LLM Agent Systems · Trustworthy AI · Federated Learning

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EDUCATION

Purdue University, Department of Computer Science *Sep 2020 – Dec 2025*

Ph.D. in Computer Science *(Advisors: Dr. Qin Hu, Dr. Snehasis Mukhopadhyay)*

Research focus: LLM Agent Systems, Agent Memory & Privacy, Federated Learning, Differential Privacy, Game Theory

Beijing Normal University, College of Information Science and Technology *Sep 2015 – Jun 2019*

B.S. in Computer Science and Technology *(GPA: 3.88 / 4.00, Rank: 6 / 55)*

PROFESSIONAL SUMMARY

Computer Science Ph.D. with 6+ years of research experience in trustworthy AI, specializing in **LLM agent systems, agent memory & privacy, and federated learning**. Author of 7+ papers in top-tier venues (IEEE TMC, IEEE TVT, ICASSP). Hands-on builder of agentic systems—including MIDAS, an agent-memory privacy framework, and Pensieve, an open-source tool for visualizing LLM agent memory. Seeking **Agent/ Applied AI Research/ Engineering** roles.

TECHNICAL SKILLS

LLM & Agent Systems: Agentic LLM design; Agent memory design (short-term, long-term, semantic); LLM pre-training & fine-tuning (SFT, LoRA); evaluation & benchmark design; RAG; MCP

AI-Native Development: Claude Code, Cursor; prompt engineering; rapid agent prototyping; AI-assisted code review

ML Research: Federated Learning, Differential Privacy, Machine Unlearning, Game Theory, Computer Vision / CNN

Languages & Tools: Python, C++, MATLAB, SQL, PyTorch, Transformers, Docker, Git, Linux, YOLO

SELECTED PROJECTS

MIDAS — Memory Integrity and Drift-Aware Safeguard for LLM Agents *NeurIPS 2026 (Submission)*

- Formalized **Context-Drift Violation (CDV)** as a new privacy primitive for agentic LLM systems; proved three formal properties, including monotonicity of privacy protection
- Built an end-to-end interactive privacy-tracing system with real-time CDV detection over agent memory; on a 2,000-instance benchmark (50 queries × 40 annotated memories), identified a **20.5% overall violation rate** spanning 0.0% (social) to 52.5% (financial), surfacing a systematic relevance–privacy tension invisible to conventional agent interfaces
- *Repository:* github.com/DrJonaC/MIDAS

Pensieve — Visualizing LLM Agent Memory *Open-Source Project, 2026*

- Designed and shipped a visualization toolkit that lets end users inspect *what an LLM agent actually knows* about them, using contextual memory compression and interactive rendering
- Enables users to gauge agent context coverage and proactively prevent privacy leakage before sharing new information with the agent
- *Repository:* github.com/DrJonaC/Pensieve

Personalized Lightweight LLM built on MiniMind *Ongoing Project, 2025*

- Built a **25M-parameter on-device AI assistant** (460 MB footprint) offering low-latency responses and one-command deployment on consumer laptops
- Added a personalization layer so the model adapts to individual user preferences; internal pilots showed noticeably more relevant answers and reduced trial-and-error for end users

Utility-Enhanced Personalized Privacy Preservation in Federated Learning *Published in IEEE TMC, 2025*

- Designed the **Group Local Differential Privacy (GLDP)** framework with a Sampling-Randomizing-Shuffling (SRS) mechanism, delivering stronger privacy without sacrificing utility
- Achieved **2–12% accuracy improvement** over SOTA across benchmarks; introduced per-participant privacy levels for real-world deployment constraints

Upcycling Noise for Federated Unlearning *Proceeding in IEEE TMC, 2025*

- Proposed the **Federated Unlearning with Indistinguishability (FUI)** framework, guaranteeing removal of client data contributions while preserving model utility

- Repurposed training-time noise to make "forget my data" requests practical, cutting the cost and disruption of full retraining

Maximizing Social Welfare in Cross-Silo Federated Learning *Published in IEEE TVT, 2024*

- Modeled cross-silo FL collaboration as a public goods game and proposed the **Multi-player Multi-action Zero-Determinant (MMZD)** strategy with an alliance variant to maximize collective utility
- Delivered stronger participation incentives, more stable training, and faster convergence for cross-organization ML partnerships

Lung Cancer Risk Stratification via Hybrid-CNN and CT Feature Decoupling *IU Health Collaboration, 2021*

- Built a CT-to-risk vision pipeline that surfaces clearer early-warning signals for clinicians; decoupled diagnostic cues from look-alike noise to reduce false alarms and sharpen patient triage

PROFESSIONAL EXPERIENCE

Research Assistant, Purdue University *Jan 2021 – Present | West Lafayette, IN*

- Led end-to-end research: problem formulation, literature-gap analysis, system design, experimental validation, writing, and publication
- Mentored **7 undergraduate students** through research projects and capstone papers

Lecturer — AIS 20000: Introduction to Data Science *Aug 2024 – Dec 2024 | West Lafayette, IN*

- Redesigned the full course for **40+ students**, integrating the latest developments in data science, ML, and big data infrastructure
- Pioneered an unconventional teaching style integrating magic tricks and variety-show formats to improve engagement; received strongly positive student feedback and renewed student curiosity toward data science

Data Operations Assistant, Beijing Didi Infinity Tech & Dev Co., Ltd. *Mar 2018 – Jun 2018 | Beijing, China*

- Designed experimental strategies evaluating the impact of coupon programs on user growth, engagement, and retention

SELECTED PUBLICATIONS

- **Chen, Jianan**, Honglu Jiang, and Qin Hu. "Utility-Enhanced Personalized Privacy Preservation in Hierarchical Federated Learning." *IEEE Transactions on Mobile Computing*, 2025. **(JCR Q1, IF 7.7, H-Index 157)**
- **Chen, Jianan**, Qin Hu, Fangtian Zhong, Yan Zhuang, and Minghui Xu. "Upcycling Noise for Federated Unlearning." *IEEE Transactions on Mobile Computing*, 2025 (Proceeding). **(JCR Q1, IF 7.7, H-Index 157)**
- **Chen, Jianan**, Qin Hu, and Honglu Jiang. "Alliance Makes Difference? Maximizing Social Welfare in Cross-Silo Federated Learning." *IEEE Transactions on Vehicular Technology*, 2023. **(JCR Q1, IF 6.7, H-Index 222)**
- **Chen, Jianan**, Qin Hu, and Honglu Jiang. "Social Welfare Maximization in Cross-Silo Federated Learning." *ICASSP 2022 — IEEE International Conference on Acoustics, Speech and Signal Processing*, IEEE, 2022.
- Chakrabarti, Subir K., **Jianan Chen**, and Qin Hu. "Stationary Markov Equilibrium Strategies in Asynchronous Stochastic Games: Existence and Computation." *Algorithms* 17.11 (2024): 490.
- **Chen, Jianan**, Qin Hu, and Honglu Jiang. "Strategic Signaling for Utility Control in Audit Games." *Computers & Security* 118 (2022): 102721.
- Peng, Cheng, Qin Hu, **Jianan Chen**, Kyubyung Kang, Feng Li, and Xukai Zou. "Energy-Efficient Device Selection in Federated Edge Learning." *2021 International Conference on Computer Communications and Networks (ICCCN)*, IEEE, 2021.

HONORS & SERVICE

Honors

- University Fellowship, Purdue University — Apr 2022
- First Prize, Beijing Undergraduate Mathematical Contest in Modeling — Sep 2017

Academic Service

- **Journal Reviewer:** IEEE Transactions on Mobile Computing (TMC), IEEE Transactions on Vehicular Technology (TVT), Journal of Network and Computer Applications (JNCA), Journal of Systems Architecture (JSA), High-Confidence Computing (HCC)
- **Conference Reviewer:** INFOCOM (2023, 2024), GLOBECOM (2022, 2023), BIBM (2024)